Fraud, Evasion, and Enforcement in Decentralized Environmental Regulation

Jacob Bradt
University of Texas at Austin

Jackson Dorsey
University of Texas at Austin

Tex-Mex Conference
UT Austin and ITAM
April 10, 2026

Delegated enforcement can create a market for fraud

Environmental regulation creates incentives for evasion

- When compliance is costly, regulated individuals may pay to avoid it

Regulators often **delegate enforcement to private third parties**

- Inspectors choose: enforce honestly, or **sell fake compliance certificates**

The result is **a market for fraud**

- **Supply:** inspectors profit by selling certificates without enforcing
- **Demand:** regulated individuals pay to avoid costly compliance

Observe delegation elsewhere: transportation licensing, food safety audits, housing inspections Examples

We study this market in vehicle emissions inspections

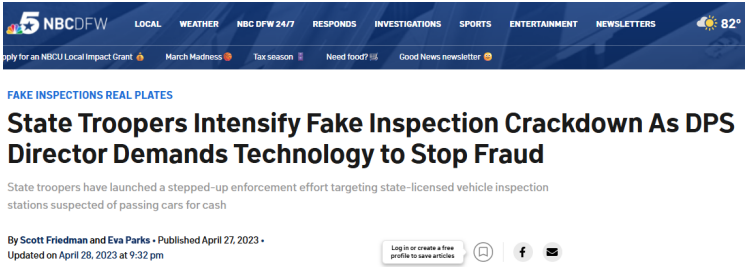
31 states + DC delegate testing to private stations

→ Drivers who fail face potentially costly repairs ($100–1000)

→ Market for fraudulent passes ("clean scans"), cost less than repair

Texas: 100M inspections across 5,500+ stations (2016–2024)

– Reports of widespread fraud → *Operation Cinderblock* (2022)



NBCDFW LOCAL WEATHER NBC DFW 24/7 RESPONDS INVESTIGATIONS SPORTS ENTERTAINMENT NEWSLETTERS 82°

pply for an NBCU Local Impact Grant March Madness Tax season Need food? Good News newsletter

FAKE INSPECTIONS REAL PLATES

State Troopers Intensify Fake Inspection Crackdown As DPS Director Demands Technology to Stop Fraud

State troopers have launched a stepped-up enforcement effort targeting state-licensed vehicle inspection stations suspected of passing cars for cash

By Scott Friedman and Eva Parks • Published April 27, 2023 • Updated on April 28, 2023 at 9:32 pm

Log in or create a free profile to save articles

Objectives

1. Quantify how much fraud exists and how much damage it causes

$\longrightarrow$ Develop a new method for detecting fraud from administrative data

- Use covariance across indicators to identify fraud rates
- Link fraud classifications to roadside emissions $\Rightarrow$ quantify externality

Objectives

1. Quantify how much fraud exists and how much damage it causes
$\longrightarrow$ Develop a new method for detecting fraud from administrative data

- Use covariance across indicators to identify fraud rates
- Link fraud classifications to roadside emissions $\Rightarrow$ quantify externality

2. Understand what drives compliance choices
$\longrightarrow$ Estimate a structural model of compliance choice

- How do drivers choose between repair, fraud, and unregistered driving?
- Exploit variation in repair costs across vehicle defects

Objectives

1. Quantify how much fraud exists and how much damage it causes
$\longrightarrow$ Develop a new method for detecting fraud from administrative data

- Use covariance across indicators to identify fraud rates
- Link fraud classifications to roadside emissions $\Rightarrow$ quantify externality

2. Understand what drives compliance choices
$\longrightarrow$ Estimate a structural model of compliance choice

- How do drivers choose between repair, fraud, and unregistered driving?
- Exploit variation in repair costs across vehicle defects

3. Evaluate whether supply-side enforcement improves outcomes
$\longrightarrow$ Evaluate counterfactual enforcement

- Predict substitution patterns when fraud access is removed
- Decompose welfare: pollution costs vs. consumer surplus vs. fraud rents

What do we find?

1. Fraud is pervasive and creates major externalities

→ 11.2 % of inspections fraudulent, peaking at 13.4 % in 2022

→ Fraud-flagged vehicles emit 2–3× excess pollution; $1.1 billion cumulative damages

What do we find?

1. Fraud is pervasive and creates major externalities

→ 11.2 % of inspections fraudulent, peaking at 13.4 % in 2022

→ Fraud-flagged vehicles emit 2–3× excess pollution; $1.1 billion cumulative damages

2. Demand for fraud is elastic and heterogeneous

→ Noncompliance rises from 2% to 68% across the repair cost distribution

→ PCV valve ($93): 2% noncompliance; catalytic converter ($750): 68%

What do we find?

1. Fraud is pervasive and creates major externalities

→ 11.2 % of inspections fraudulent, peaking at 13.4 % in 2022

→ Fraud-flagged vehicles emit 2–3× excess pollution; $1.1 billion cumulative damages

2. Demand for fraud is elastic and heterogeneous

→ Noncompliance rises from 2% to 68% across the repair cost distribution

→ PCV valve ($93): 2% noncompliance; catalytic converter ($750): 68%

3. Enforcement generates leakage

→ Targeting fraud stations: ↓ fraud but ↑ unregistered driving

→ Fraud rents exceed pollution gains—helps explain why fraud persists

Related literature

- **Detecting fraud & misconduct:** DellaVigna and La Ferrara, 2010; Fisman and Wei, 2004; Ghanem and Zhang, 2014; Jacob and Levitt, 2003 . . .
  - → *Cross-indicator covariance identifies the level of fraud*
- **Illegal markets & delegated enforcement:**
  - Theory: Becker and Stigler, 1974; Laffont and Tirole, 1993; Shleifer and Vishny, 1993 . . .
  - Empirics: Bennett et al., 2013; Duflo et al., 2013, 2018; Greenstone et al., 2022; Habte and Holm, 2022; Oliva, 2015; Olken, 2007 . . .
  - → *We trace enforcement through demand responses to welfare*
- **Environmental enforcement & vehicle emissions:** Ale-Chilet et al., 2025; Blundell et al., 2020; Jacobsen et al., 2023; Reynaert and Sallee, 2021; Shimshack and Ward, 2005; Zou, 2021 . . .
  - → *We document enforcement leakage across noncompliance margins*

Outline

1 Introduction

2 Background and Data

3 Fraud Detection Model

4 Descriptive Evidence

5 Demand for Fraud

6 Counterfactuals

7 Conclusion

The Texas Inspection and Maintenance Program

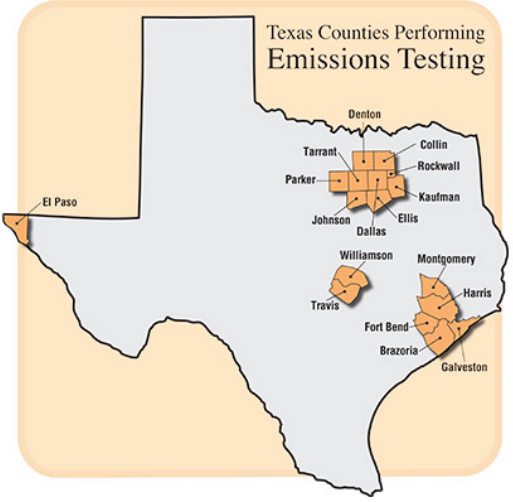
17 CAA non-attainment counties

- DFW, Houston, Austin, El Paso metro areas
- Annual OBD-II test for post-1996 vehicles

The inspection market

- 5,500+ stations, mostly independent auto shops
- Low margins on testing ($18.50 regulated fee)

→ Stations enter to attract repair customers



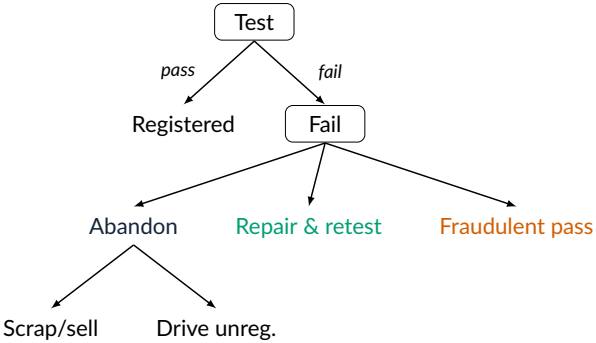
The driver's problem

Annual cycle

- Registration requires passing emissions test
- Most pass; if fail, driver learns the specific defect

Repair costs vary widely:

O_2 sensor	~$75
Spark plugs	~$275
Catalytic converter	~$600+



How fraud works: “Clean scanning”

Standard inspection: Analyzer connects to vehicle and downloads

1. **Electronic identity:** eVIN, calibration IDs, PCM identifier, . . .
2. **Readiness monitors:** status of self-diagnostic tests
3. **DTCs & MIL:** trouble codes and check engine light status

Two methods to bypass this:

1. **OBD Simulator:** Device mimics a passing OBD system
2. **Surrogate vehicle:** Test a clean car, attribute results to failing vehicle

Vehicle need not be present → enables “assembly line” fraud

Both methods leave **electronic fingerprints** in the data

→ We also observe real-world tailpipe emissions from the **TORVET** roadside sensing program, linked to inspection records via license plate

Fraud indicators

We construct **5 indicator groups** from the electronic inspection data:

Throughput	Test completed <5 min after previous test on same analyzer

Detailed definitions

Fraud indicators

We construct **5 indicator groups** from the electronic inspection data:

Throughput	Test completed <5 min after previous test on same analyzer
eVIN mismatch	Electronic VIN missing or doesn't match inspector-entered VIN

Detailed definitions

Fraud indicators

We construct **5 indicator groups** from the electronic inspection data:

Throughput	Test completed <5 min after previous test on same analyzer
eVIN mismatch	Electronic VIN missing or doesn't match inspector-entered VIN
Profile recycling	Electronic profile identical to previous test on same analyzer

Detailed definitions

Fraud indicators

We construct **5 indicator groups** from the electronic inspection data:

Throughput	Test completed <5 min after previous test on same analyzer
eVIN mismatch	Electronic VIN missing or doesn't match inspector-entered VIN
Profile recycling	Electronic profile identical to previous test on same analyzer
Manual entry	VIN typed manually instead of barcode scanned

Detailed definitions

Fraud indicators

We construct **5 indicator groups** from the electronic inspection data:

Throughput	Test completed <5 min after previous test on same analyzer
eVIN mismatch	Electronic VIN missing or doesn't match inspector-entered VIN
Profile recycling	Electronic profile identical to previous test on same analyzer
Manual entry	VIN typed manually instead of barcode scanned
Protocol mismatch	OBD communication protocol inconsistent with vehicle model

Detailed definitions

Fraud indicators

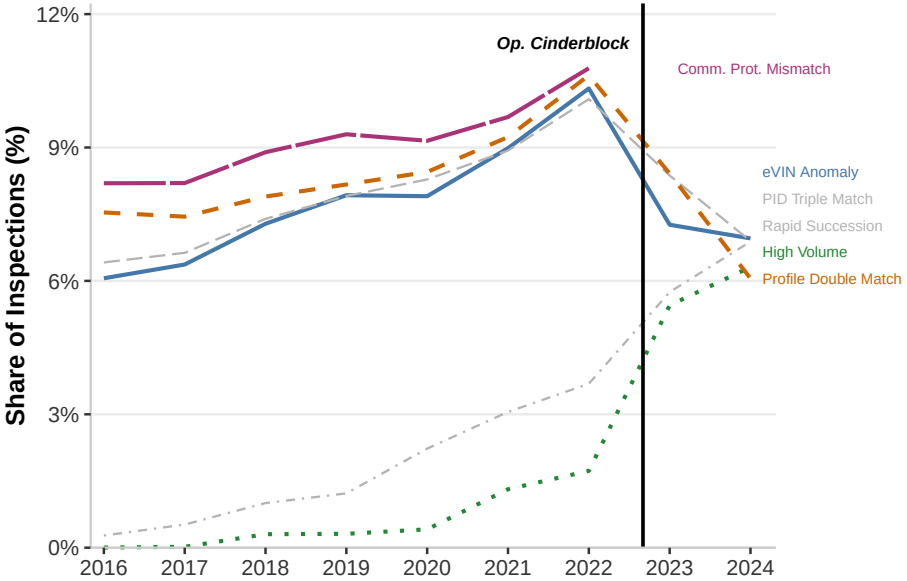
We construct **5 indicator groups** from the electronic inspection data:

Throughput	Test completed <5 min after previous test on same analyzer
eVIN mismatch	Electronic VIN missing or doesn't match inspector-entered VIN
Profile recycling	Electronic profile identical to previous test on same analyzer
Manual entry	VIN typed manually instead of barcode scanned
Protocol mismatch	OBD communication protocol inconsistent with vehicle model

Each group draws on a **different forensic dimension** of the inspection record

Detailed definitions

Fraud indicators rise through 2022, then fall sharply



Outline

1 Introduction

2 Background and Data

3 Fraud Detection Model

4 Descriptive Evidence

5 Demand for Fraud

6 Counterfactuals

7 Conclusion

Detecting fraud with statistical indicators

Standard approach:

- Construct a statistical indicator of misconduct
  - E.g., suspicious answer patterns identify teacher cheating (Jacob and Levitt, 2003)
- Show it is elevated in suspicious subsamples

→ Establishes that fraud *exists*—but difficult to identify *how much*

Detecting fraud with statistical indicators

Standard approach:

- Construct a statistical indicator of misconduct
  - E.g., suspicious answer patterns identify teacher cheating (Jacob and Levitt, 2003)
- Show it is elevated in suspicious subsamples

→ Establishes that fraud *exists*—but difficult to identify *how much*

Why not? An indicator can fire for two reasons:

$$\underbrace{p_j}_{\substack{\text{observed}\\ \text{flag rate}}} = \underbrace{\pi}_{\text{Prob(Fraud)}} \cdot \underbrace{\beta_j}_{\substack{\text{P(flag|fraud)}\\ \text{detection rate}}} + (1-\pi) \cdot \underbrace{\Lambda_j}_{\substack{\text{P(flag|honest)}\\ \text{false positive rate}}}$$

Detecting fraud with statistical indicators

Standard approach:

- Construct a statistical indicator of misconduct
  - E.g., suspicious answer patterns identify teacher cheating (Jacob and Levitt, 2003)
- Show it is elevated in suspicious subsamples

→ Establishes that fraud *exists*—but difficult to identify *how much*

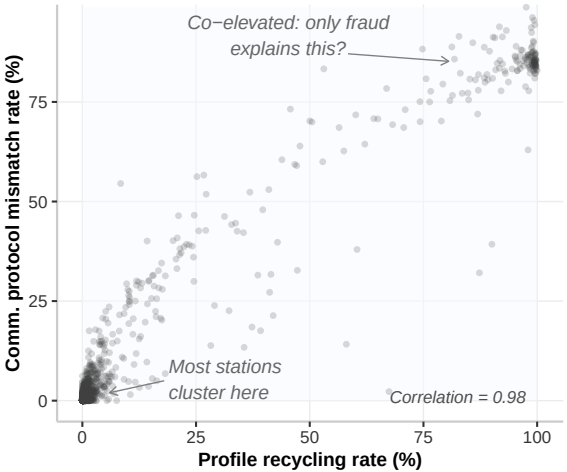
Why not? An indicator can fire for two reasons:

$$\underbrace{p_j}_{\substack{\text{observed}\\ \text{flag rate}}} = \underbrace{\pi}_{\text{Prob(Fraud)}} \cdot \underbrace{\beta_j}_{\substack{\text{P(flag|fraud)}\\ \text{detection rate}}} + (1-\pi) \cdot \underbrace{\Lambda_j}_{\substack{\text{P(flag|honest)}\\ \text{false positive rate}}}$$

Example: 8% of inspections at a station have mismatched VINs

- $\pi = 10\%, \beta = 80\%, \Lambda = 0\%? \rightarrow p = 8\%$ (fraud, high detection)
- $\pi = 0\%, \Lambda = 8\%? \rightarrow p = 8\%$ (no fraud, noisy indicator)

Our approach: Cross-indicator covariance



Data pattern: Two indicators from *different* data streams, plotted at the station level

- Cluster near zero: Cov $= 0$ $\rightarrow$ honest stations
- Co-elevated cluster: Cov $> 0 \rightarrow$ fraud stations

Implication: No mechanical link; only latent cause (fraud) generates correlation

$\Rightarrow$ Magnitude of covariance identifies π

The key equation

Proposition: If fraud indicators are conditionally independent given fraud status, the covariance between any two indicators j and k within a cell is:

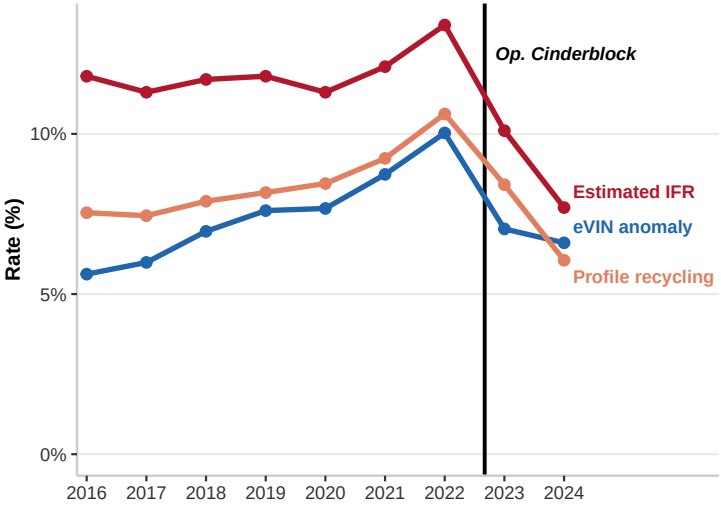
$$\mathrm{Cov}(Y_j, Y_k \mid s, v) \;=\; \pi_{sv}(1 - \pi_{sv}) \cdot D_j \cdot D_k$$

- Detection gap: $D_j = \beta_j - \Lambda_j$
  - $\rightarrow$ *How much more indicator j fires on fraud vs. honest inspections*
- Excess flag rate: $\delta_j = p_j - \Lambda_j = \pi \cdot D_j$
  - $\rightarrow$ *The fraud-attributable portion of the flag rate*

In practice, we use $K = 5$ indicators $\rightarrow$ 31 moments, 11 parameters, **20 overidentifying restrictions**

EM algorithm | When does CI fail? | MC frontier | MC validation

Estimated fraud rate: 11.2 % (2016–2024)



Scale: 10.7 million cumulative fraudulent inspections

Concentration: 660 high-intensity stations drive most fraud volume

Why a model? Estimated inspection fraud rate (IFR) exceeds each indicator

IFR by metro | Volume | 2022 table | Validation | Station classifications

Outline

1 Introduction

2 Background and Data

3 Fraud Detection Model

4 Descriptive Evidence

5 Demand for Fraud

6 Counterfactuals

7 Conclusion

Fact 1: Fraud-flagged vehicles emit ∼3× more pollution

Match TORVET roadside captures to most recent inspection:

$$\underbrace{y_{ivt}}_{\substack{\text{emissions}\\ \text{(g/kg-fuel)}}} = \underbrace{\beta_1 \cdot \text{Fraud}_i}_{\substack{\hat{\tau}_i > 0.5\\ \text{from model}}} + \underbrace{\beta_2 \cdot \text{Unreg}_i}_{\substack{\text{registration}\\ \text{lapsed} \geq 6\text{mo}}} + \underbrace{\gamma \mathbf{X}_{ivt}}_{\substack{\text{speed, accel.,}\\ \text{VSP}}} + \underbrace{\delta_{\text{make}} + \delta_{\text{MY}} + \delta_{\text{date}}}_{\substack{\text{make, model-year,}\\ \text{capture-date FEs}}} + \varepsilon_{ivt}$$

Fact 1: Fraud-flagged vehicles emit ~3× more pollution

Match TORVET roadside captures to most recent inspection:

$$\underbrace{y_{ivt}}_{\substack{\text{emissions}\\ \text{(g/kg-fuel)}}} = \underbrace{\beta_1 \cdot \text{Fraud}_i}_{\substack{\hat{\tau}_i>0.5\\ \text{from model}}} + \underbrace{\beta_2 \cdot \text{Unreg}_i}_{\substack{\text{registration}\\ \text{lapsed}\geq 6\text{mo}}} + \underbrace{\gamma \mathbf{X}_{ivt}}_{\substack{\text{speed, accel.,}\\ \text{VSP}}} + \underbrace{\delta_{\text{make}} + \delta_{\text{MY}} + \delta_{\text{date}}}_{\substack{\text{make, model-year,}\\ \text{capture-date FEs}}} + \varepsilon_{ivt}$$

	NO (g/kg)	HC (g/kg)
Fraud	2.23	0.45
	(0.03)	(0.01)
Unreg.	0.94	0.42
	(0.03)	(0.02)
Control	1.09	0.19

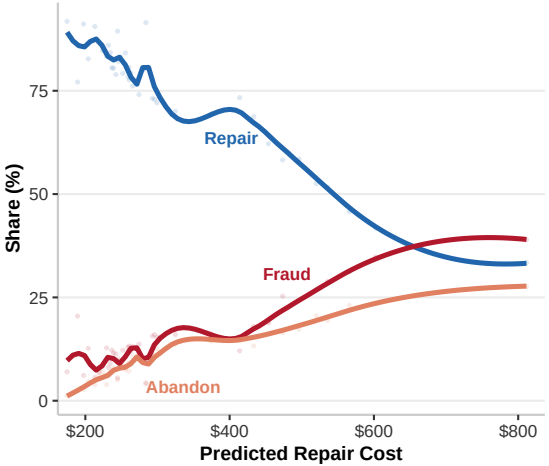
Health damages (AP4):

- **$121 /yr** per fraud vehicle
- **$1.1 billion** cumulative (2017–2024)

Note: Accounting exercise—not a policy counterfactual

Damages over time Full table

Fact 2: Noncompliance responds sharply to repair costs



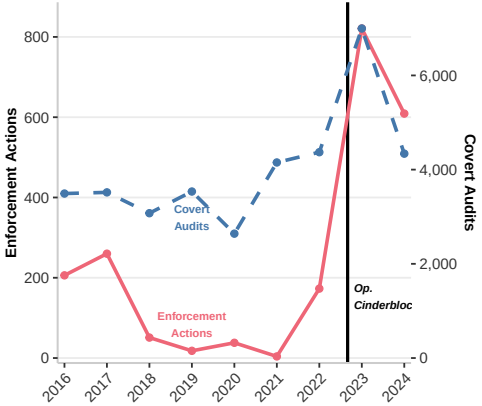
Predicted repair cost: uses OBD diagnostic data

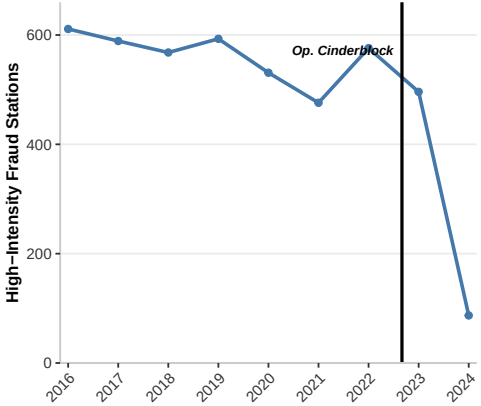
→ Conditional on make, station, year FEs: variation from *specific defect*

Key finding: cost drives comply vs. non-comply

Motivates model to predict substitution under enforcement counterfactuals

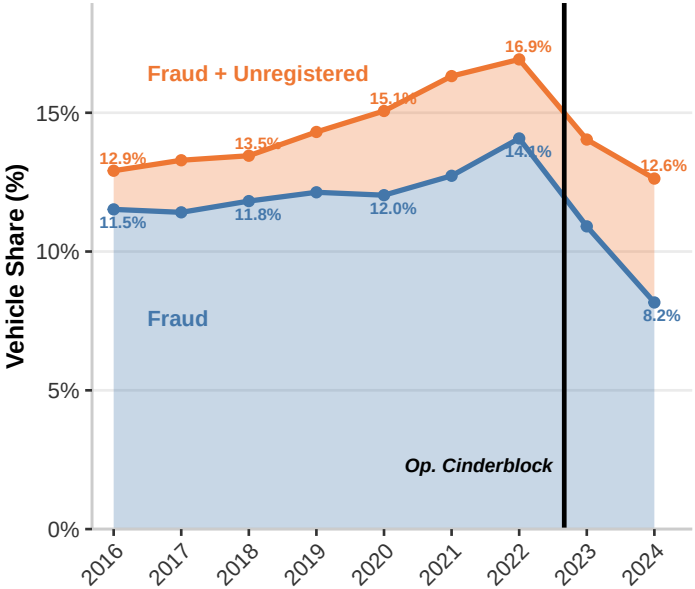
Fact 3: “Operation Cinderblock” shut down fraud supply





Post-Cinderblock: enforcement actions rose to 821 (2023); high-intensity fraud stations fell from 13% to 2%

Fact 4: Leakage offset 1/3 of "Cinderblock" effect



Fraud ↓~6pp

Unregistered ↑~2pp

⇒ Combined noncompliance fell only ~4pp

Outline

1 Introduction

2 Background and Data

3 Fraud Detection Model

4 Descriptive Evidence

5 Demand for Fraud

6 Counterfactuals

7 Conclusion

Compliance choice model

Vehicles that fail choose **repair or not**; if not, they choose **fraud or abandon**:

Top level — Repair vs. Non-comply:

$$V_{\text{repair},i} = \mathbf{x}_i'\boldsymbol{\beta}_r + \alpha_i \cdot r_i \qquad\qquad V_{\text{nc},i} = \mathbf{x}_i'\boldsymbol{\beta}_{\text{nc}} + \text{IV}_i$$

- $\mathbf{x}_i$: vehicle age bins, county FEs, quadratic time trend
- r_i: predicted repair cost (from OBD diagnostics)
- $\alpha_i = -\exp(\mu + \sigma\nu_i), \quad \nu_i \sim N(0,1)$ (log-normal, always < 0)
- IV_i: inclusive value summarizing value of non-comply options

Compliance choice model

Vehicles that fail choose **repair or not**; if not, they choose **fraud or abandon**:

Top level — Repair vs. Non-comply:

$$V_{\text{repair},i} = \mathbf{x}_i'\boldsymbol{\beta}_r + \alpha_i \cdot r_i \qquad\qquad V_{\text{nc},i} = \mathbf{x}_i'\boldsymbol{\beta}_{\text{nc}} + \text{IV}_i$$

- $\mathbf{x}_i$: vehicle age bins, county FEs, quadratic time trend
- r_i: predicted repair cost (from OBD diagnostics)
- $\alpha_i = -\exp(\mu + \sigma\nu_i), \quad \nu_i \sim N(0,1)$ (log-normal, always < 0)
- IV_i: inclusive value summarizing value of non-comply options

Within nest — Fraud vs. Abandon:

- **Fraud:** Obtain fraudulent pass at one of available stations (no repair needed)
- **Abandon:** Exit inspection system (unregistered driving or scrap/sell)

The inclusive value: What's inside the non-comply nest

$V_{\text{nc},i} = \mathbf{x}_i'\boldsymbol{\beta}_{\text{nc}} + \text{IV}_i$, where:

$$\text{IV}_i = \lambda \cdot \log\Big(\underbrace{N_{\text{opp}}\, e^{\delta_{\text{opp}}/\lambda} + N_{\text{hi}}\, e^{\delta_{\text{hi}}/\lambda} + N_{\text{ho}}\, e^{\delta_{\text{ho}}/\lambda}}_{\text{fraud stations (3 types)}} + \underbrace{e^{\mathbf{z}_i'\boldsymbol{\beta}_a/\lambda}}_{\text{abandon}} \Big)$$

Three fraud station types (from fraud detection model):

- N_{hi}: High-intensity in-county (dedicated fraud shops, $\hat{\pi}_s > 25\%$)
- N_{opp}: Opportunistic in-county (occasional fraud, not primary business)
- N_{ho}: High-intensity out-of-county (remote clean scanning)

δ_k: utility of fraud at each type, *net of equilibrium bribe price*

Nesting parameter $\lambda \in (0, 1)$: governs within-nest substitution

- $\lambda \to 0$: new fraud station cannibalizes other fraud stations
- $\lambda \to 1$: new station draws equally from repair and abandon

Nesting diagram

Identification

Repair cost sensitivity (μ, σ):

- Variation in predicted repair costs from diagnostics
- μ: average fraud rate gradient across repair cost distribution
- σ: curvature of gradient (how response varies across distribution)

Nesting parameter (λ):

- $N_{k,mt}$ varies across counties and over time
- Quasiexperimental variation from Operation Cinderblock
- Does removing stations shift drivers to repair or to abandon?

Station type utilities (δ_k):

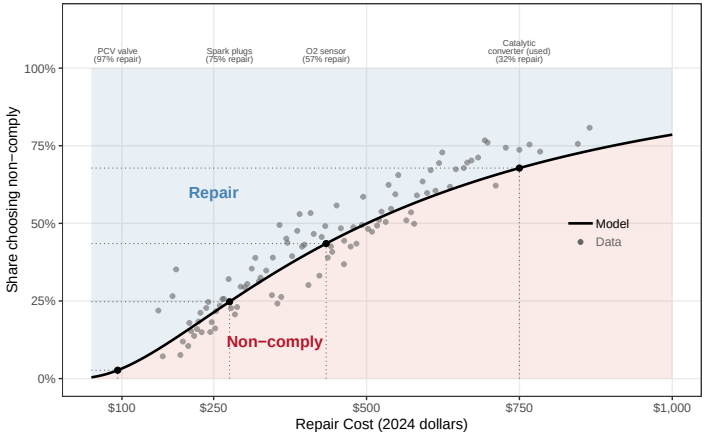
- Relative choice probabilities across station types, conditional on total stations

Estimation results

	Estimate	Std. Error
Repair cost sensitivity (log-normal)		
μ (location)	0.329	0.075
σ (scale)	0.874	0.008
$\mathbb{E}[\alpha]$ (mean, per \$100)	−2.04	0.167
Nesting		
λ (nesting parameter)	0.857	0.017
Station type utilities (within nest)		
δ_{opp} (opportunistic)	-6.253	0.142
δ_{hi} (high-intensity, in-county)	-5.224	0.108
δ_{ho} (high-intensity, out-of-county)	-6.184	0.085
Implied elasticities (at mean repair cost)		
ε(repair, RC)	1.19	
ε(non-comply, RC)	4.41	
ε(repair, N_{hi})	-0.0424	

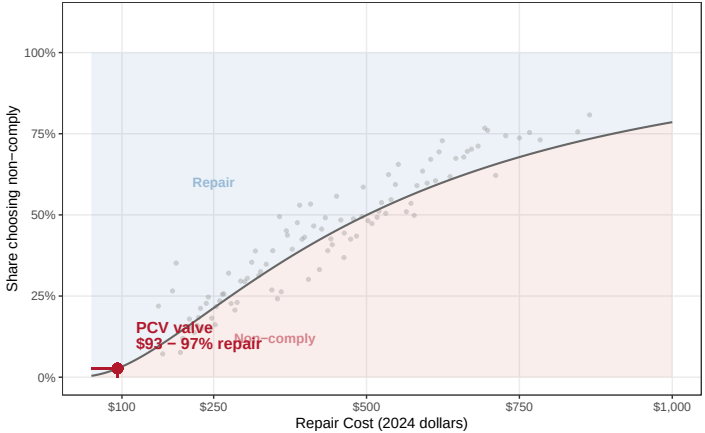
- $\hat{\lambda}$: fraud and abandon are closer substitutes than either is to repair
- Large $\hat{\sigma}$: substantial heterogeneity in cost sensitivity
- Elastic demand: noncompliance responsive to both repair costs and station availability

Indifference repair costs



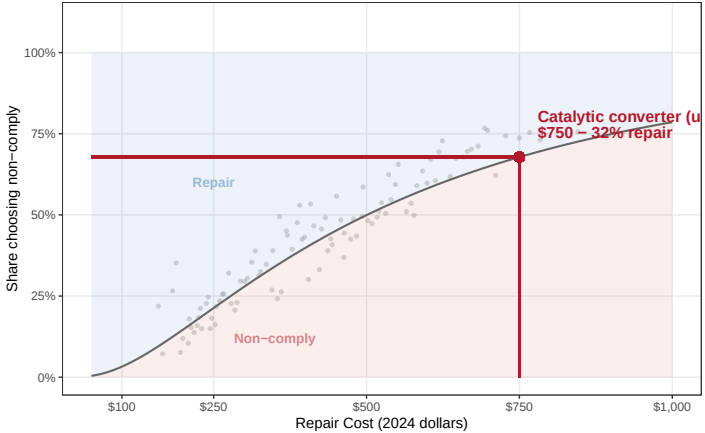
Median driver switches to noncompliance at $501

Indifference repair costs



At $93 (PCV valve): 97% repair—cheap fixes generate almost no noncompliance

Indifference repair costs



At $750 (catalytic converter): only 32% repair—expensive fixes push most into noncompliance

Outline

1 Introduction

2 Background and Data

3 Fraud Detection Model

4 Descriptive Evidence

5 Demand for Fraud

6 Counterfactuals

7 Conclusion

Evaluating supply-side enforcement

Use the demand model to predict what happens when fraud stations are removed (522,946 initial-fail vehicles, 2022):

1. **Remove high-intensity stations** ($N_{\text{hi}} = N_{\text{ho}} = 0$)
   → Mimics targeted enforcement (Operation Cinderblock)

2. **Remove all fraud stations** ($N_{\text{opp}} = N_{\text{hi}} = N_{\text{ho}} = 0$)
   → Benchmark: e.g., California-style electronic VIN verification

Abandon decomposition / bounding

Substitution patterns: enforcement leakage

	Baseline	Remove High-Intensity	Remove All Fraud
Repair (%)	78.9	81.2	82.7
Fraud (%)	13.6	6.4	0.0
Abandon (%)	7.5	12.4	17.3
Unregistered (%)	3.9	6.4–8.7	9.0–13.7
Fraud + Unreg (%)	**17.4**	**12.8–15.1**	**9.0–13.7**

- Not all displaced fraud converts to repair—many substitute to abandon
- Unregistered bounds: lower assumes all marginal abandoners drive unreg.; upper uses estimated $P(\text{unreg} \mid \text{abandon})$

Welfare decomposition

	Remove High-Int.	Remove All
A: Drivers & repair		
ΔCS (drivers)	$-5.25	$-8.64
ΔPS repair sector	$1.58	$2.58
B: Pollution benefits		
Lower bound	$2.71	$4.63
Upper bound	$5.33	$10.21
C: Fraud station surplus		
ΔPS fraud stations	$-10.76	$-20.33
D: Net excl. fraud PS		
Lower bound	$-0.96	$-1.43
Upper bound	$1.66	$4.15
E: Net incl. fraud PS		
Lower bound	$-11.73	$-21.76
Upper bound	$-9.10	$-16.18

Per initial-fail driver, 2022

Drivers lose surplus when fraud is removed

- Forced into less-preferred alternatives
- Repair sector gains partially offset

Welfare decomposition

	Remove High-Int.	Remove All
A: Drivers & repair		
ΔCS (drivers)	\$-5.25	\$-8.64
ΔPS repair sector	\$1.58	\$2.58
B: Pollution benefits		
Lower bound	\$2.71	\$4.63
Upper bound	\$5.33	\$10.21
C: Fraud station surplus		
ΔPS fraud stations	\$-10.76	\$-20.33
D: Net excl. fraud PS		
Lower bound	\$-0.96	\$-1.43
Upper bound	\$1.66	\$4.15
E: Net incl. fraud PS		
Lower bound	\$-11.73	\$-21.76
Upper bound	\$-9.10	\$-16.18

Per initial-fail driver, 2022

Pollution benefits exceed CS losses

- Panel D: enforcement is welfare-improving under upper bound (excluding fraud rents)
- The case for enforcement looks strong…

Welfare decomposition

	Remove High-Int.	Remove All
A: Drivers & repair		
ΔCS (drivers)	\$-5.25	\$-8.64
ΔPS repair sector	\$1.58	\$2.58
B: Pollution benefits		
Lower bound	\$2.71	\$4.63
Upper bound	\$5.33	\$10.21
C: Fraud station surplus		
ΔPS fraud stations	\$-10.76	\$-20.33
D: Net excl. fraud PS		
Lower bound	\$-0.96	\$-1.43
Upper bound	\$1.66	\$4.15
E: Net incl. fraud PS		
Lower bound	\$-11.73	\$-21.76
Upper bound	\$-9.10	\$-16.18

Per initial-fail driver, 2022

But fraud rents are large

- Stations earn ~\$150/inspection at near-100% margins
- Rents destroyed dwarf pollution gains
- Panel E: net welfare *loss* under all assumptions

Whether fraud profits count is normative—but their *magnitude* helps explain persistence

Delegation creates a market for fraud

Why fraud persists:

- Fraud rents exceed environmental gains from enforcement
- This surplus finances investment in sustaining the market

Supply-side enforcement alone is insufficient:

- Enforcement leakage: displaced drivers substitute to unregistered driving, not repair

Effective enforcement requires complementary demand-side tools:

- Registration enforcement to close the unregistered margin
- Repair subsidies to shift displaced drivers toward compliance
- Subsidies for scrapage (“cash for clunkers”)

Outline

1 Introduction

2 Background and Data

3 Fraud Detection Model

4 Descriptive Evidence

5 Demand for Fraud

6 Counterfactuals

7 Conclusion

Summary

1. **Detection:** Cross-indicator covariance $\rightarrow$ 11.2 % of inspections fraudulent
2. **Damages:** $\sim 3\times$ excess pollution; \$1.1 billion cumulative health damages
3. **Demand:** Median driver switches to noncompliance at \$501
4. **Leakage:** Enforcement reduces fraud but $\sim 1/3$ of displaced drivers go unregistered

Policy: Supply-side enforcement alone is insufficient—need complementary demand-side tools

Thank you!

Please reach out with comments or questions:

Jacob Bradt: jacob.bradt@mccombs.utexas.edu

Jackson Dorsey: jackson.dorsey@austin.utexas.edu

References I

Ale-Chilet, J., Chen, C., Li, J., & Reynaert, M. (2025).Colluding Against Environmental Regulation. *The Review of Economic Studies*, rdaf024.

Becker, G. S., & Stigler, G. J. (1974).Law Enforcement, Malfeasance, and Compensation of Enforcers. *The Journal of Legal Studies*, *3*(1), 1–18.

Bennett, V. M., Pierce, L., Snyder, J. A., & Toffel, M. W. (2013).Customer-Driven Misconduct: How Competition Corrupts Business Practices. *Management Science*, *59*(8), 1725–1742.

Blundell, W., Gowrisankaran, G., & Langer, A. (2020).Escalation of Scrutiny: The Gains from Dynamic Enforcement of Environmental Regulations. *American Economic Review*, *110*(8), 2558–2585.

DellaVigna, S., & La Ferrara, E. (2010).Detecting Illegal Arms Trade. *American Economic Journal: Economic Policy*, *2*(4), 26–57.

Duflo, E., Greenstone, M., Pande, R., & Ryan, N. (2013).Truth-telling by Third-party Auditors and the Response of Polluting Firms: Experimental Evidence from India*. *The Quarterly Journal of Economics*, *128*(4), 1499–1545.

Duflo, E., Greenstone, M., Pande, R., & Ryan, N. (2018).The Value of Regulatory Discretion: Estimates From Environmental Inspections in India. *Econometrica*, *86*(6), 2123–2160.

Fisman, R., & Wei, S.-J. (2004).Tax Rates and Tax Evasion: Evidence from "Missing Imports" in China. *Journal of Political Economy*, *112*(2), 471–496.

Ghanem, D., & Zhang, J. (2014).Effortless Perfection: Do Chinese cities manipulate air pollution data? *Journal of Environmental Economics and Management*, *68*(2), 203–225.

Greenstone, M., He, G., Jia, R., & Liu, T. (2022).Can Technology Solve the Principal-Agent Problem? Evidence from China's War on Air Pollution. *American Economic Review: Insights*, *4*(1), 54–70.

Habte, O. A., & Holm, H. J. (2022).Competition makes inspectors more lenient: Evidence from the motor vehicle inspection market. *Review of Industrial Organization*, *61*(1), 45–72.

Jacob, B. A., & Levitt, S. D. (2003).Rotten Apples: An Investigation of the Prevalence and Predictors of Teacher Cheating*. *The Quarterly Journal of Economics*, *118*(3), 843–877.

Jacobsen, M. R., Sallee, J. M., Shapiro, J. S., & van Benthem, A. A. (2023).Regulating Untaxable Externalities: Are Vehicle Air Pollution Standards Effective and Efficient?*. *The Quarterly Journal of Economics*, *138*(3), 1907–1976.

Laffont, J.-J., & Tirole, J. (1993). *A theory of incentives in procurement and regulation*. MIT Press.

Oliva, P. (2015).Environmental regulations and corruption: Automobile emissions inspections in Mexico City. *Journal of Political Economy*, *123*(3), 686–723.

Olken, B. A. (2007).Monitoring Corruption: Evidence from a Field Experiment in Indonesia. *Journal of Political Economy*, *115*(2), 200–249.

Reynaert, M., & Sallee, J. M. (2021).Who Benefits When Firms Game Corrective Policies? *American Economic Journal: Economic Policy*, *13*(1), 372–412.

Shimshack, J. P., & Ward, M. B. (2005).Regulator reputation, enforcement, and environmental compliance. *Journal of Environmental Economics and Management*, *50*(3), 519–540.

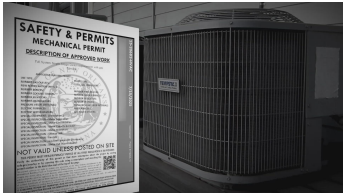
Shleifer, A., & Vishny, R. W. (1993).Corruption*. *The Quarterly Journal of Economics*, *108*(3), 599–617.

Zou, E. Y. (2021).Unwatched Pollution: The Effect of Intermittent Monitoring on Air Quality. *American Economic Review*, *111*(7), 2101–2126.

Appendix: Delegated inspections gone wrong

Transportation: Private driving schools sold fake commercial licenses; 6,000+ fraudulent CDLs, 13+ deaths — USA (2025)

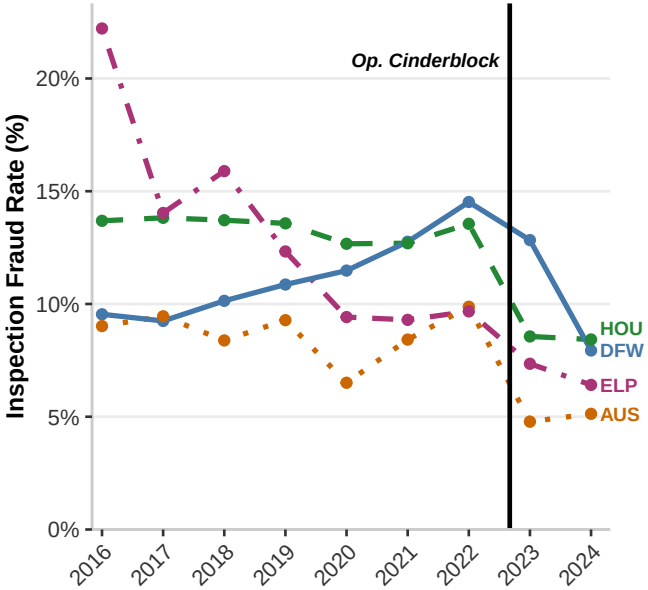
Food safety: Private auditor gave "superior" rating to plant with mold and *Salmonella*; 9 deaths, 700+ sickened — Peanut Corp. of America (2009)



Housing: Private inspectors reused photos from other properties to fake reports; owner charged with bribing the mayor — New Orleans (2021)

Back

Appendix: Inspection fraud rate by metro area

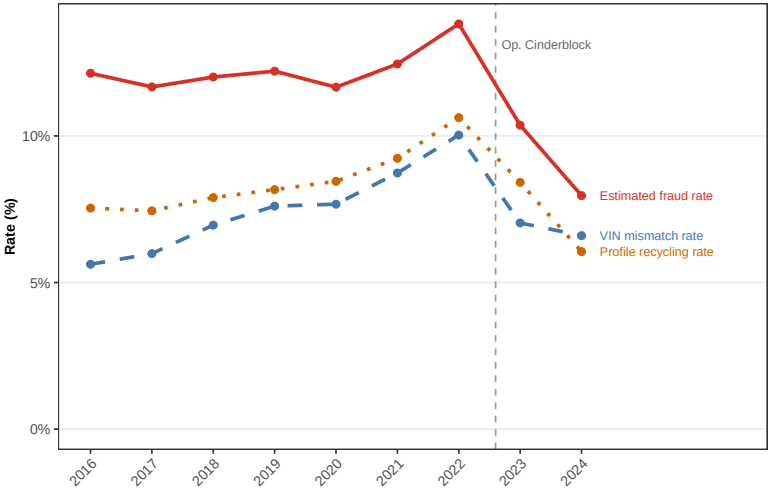


Appendix: Classification results (2022)

	Stations			Inspections		
Region	Total	Fraud (%)	High-int.	Total	Fraudulent	IFR (%)
Dallas–Fort Worth	2,416	1,483 (61)	212	5.1M	747K	14.5
Houston	2,312	1,585 (69)	364	4.4M	597K	13.6
Austin	452	334 (74)	69	1.2M	119K	9.9
El Paso	239	159 (67)	15	565K	55K	9.7
Total	**5,419**	**3,561 (66)**	**660**	**11.3M**	**1.5M**	**13.4**

Back to results

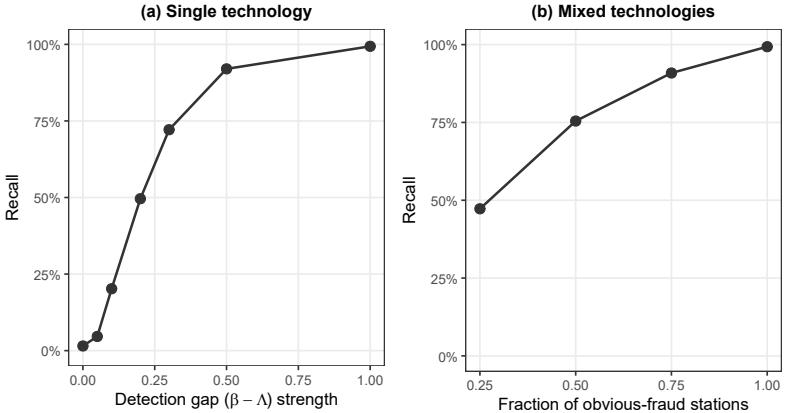
Appendix: Estimated fraud rate vs. raw indicators



- Estimated fraud rate is higher than any individual indicator
  - eVIN misses older vehicles (pre-2005 never return an eVIN)
  - Profile recycling misses fraud when inspectors swap surrogates

Back to results

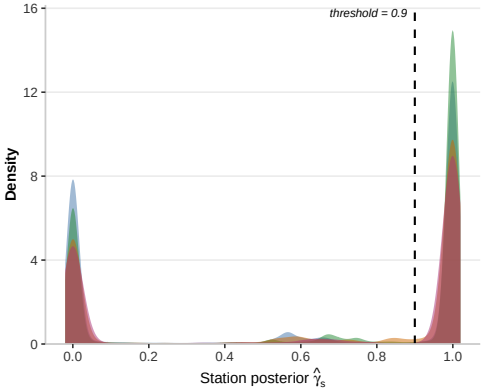
Appendix: Monte Carlo identification frontier



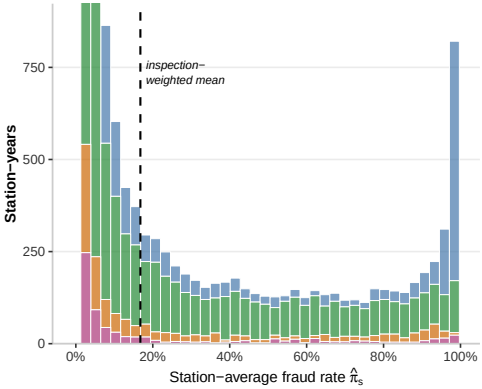
Recall degrades smoothly as detection gaps shrink ($\kappa \to 0$). With $K = 5$ indicators, the estimator maintains >90% recall down to $\kappa \approx 0.3$.

Back to model

Appendix: Station classification and fraud rate distributions



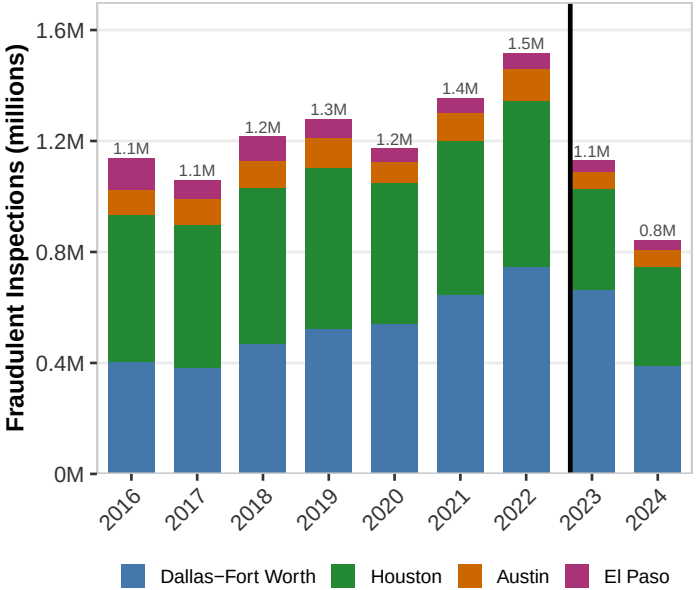
Posterior $\hat{\gamma}_s$ is sharply bimodal: stations are clearly clean or fraud-capable



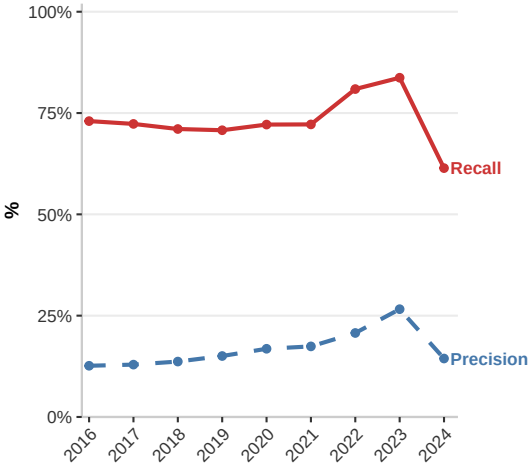
Among fraud-classified stations, fraud rates vary widely (median 1.2 %, 90th pctile 63.9 %)

Back to results

Appendix: Fraudulent inspections over time



Appendix: External validation



- 94% of sanctioned stations classified as fraud-capable
- Monte Carlo: 99% recall, 94% precision

Back to results

Appendix: Fraud indicator definitions

1. **Electronic profile mismatch**
   - CAL_ID, CVN, PCM_ID, or PID_COUNT changes between tests for same vehicle
   - Suggests surrogate vehicle or OBD simulator used
2. **Missing OBD_VIN**
   - Model year 2005+ vehicles must report e-VIN during OBD inspection
   - Missing e-VIN suggests device not connected to actual vehicle
3. **Outside business hours**
   - Tests conducted before 6:00 AM or after 9:00 PM
   - Unusual hours associated with fraudulent operations

Back

Appendix: Fraud indicator definitions (cont.)

4. Excessive DTCs with pass

- Passing test despite >1 stored diagnostic trouble code (MY > 2000)
- Indicates potential manipulation of test result

5. Excessive monitor unreadiness

- Passing test with ≥2 readiness monitors not ready (MY > 2000)
- Suggests emissions system was recently reset or bypassed

Back

Appendix: Why EM? Identification vs. estimation

Identification: Covariance identifies π_{sv} in closed form—but requires honest baselines Λ_j, which require knowing which stations are clean

Chicken-and-egg: To estimate π, need Λ; to estimate Λ, need to classify stations (i.e., know something about π)

EM solves simultaneously:

- **E-step:** Compute posteriors γ_s (station type) and τ_i (inspection fraud status)
- **M-step:** Re-estimate Λ_j, β_j, π_{sv} using posteriors as weights
- Iterate; classify stations with $\gamma_s > 0.9$

31 moments per cell, 11 parameters: heavily overidentified. MC: 99% recall, 94% precision, RMSE 0.039.

Back to model

Appendix: When does conditional independence fail?

Assumption: Indicators fire independently conditional on fraud status

This can fail when indicators are mechanically related:

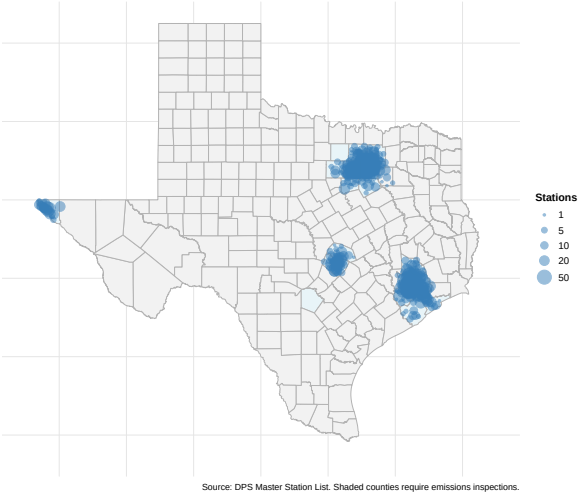
- “>5 tests in 5 minutes” and “>3 tests in 3 minutes” overlap by construction
- A station that triggers one throughput indicator almost certainly triggers the other
- Creates positive covariance *even at honest stations*, violating CI

Our approach:

- Construct indicators measuring *different aspects* of the inspection: eVIN identity, throughput, profile recycling, communication protocol
- Avoid mechanically overlapping indicators within the same forensic dimension
- With $K \geq 4$ indicators, CI is testable via overidentifying restrictions

Back

Appendix: Station map



Stations active in 2024 plotted at zip centroid; circle size $\propto$ stations per zip

Appendix: Bounding the unregistered driving share

Problem: "Abandon" combines unregistered driving (pollution) and exit (no pollution)

Solution: TORVET roadside sensors capture vehicles regardless of registration status

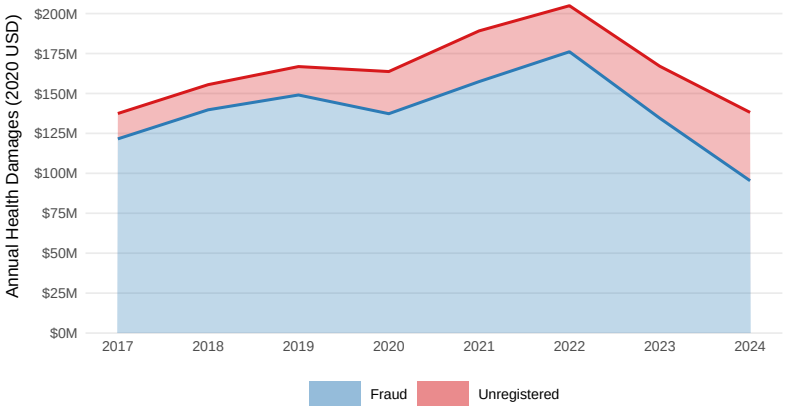
$$P(\text{unreg} \mid \text{abandon}, \mathbf{x}_i) = \frac{P(\text{in TORVET}_{t+1} \mid \text{abandon}, \mathbf{x}_i)}{P(\text{in TORVET}_{t+1} \mid \text{registered}, \mathbf{x}_i)}$$

Bounds for counterfactuals:

- *Lower bound* on pollution gains: all marginal abandon-switchers drive unregistered
- *Upper bound:* marginal switchers have same $P(\text{unreg} \mid \text{abandon})$ as current abandoners

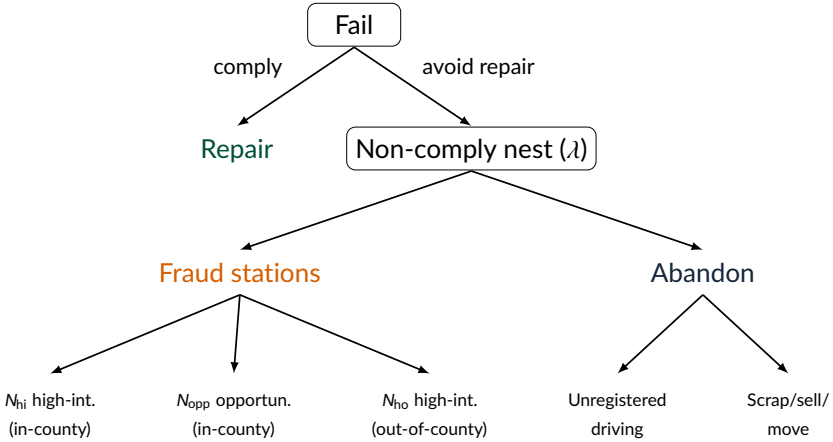
Back

Appendix: Health damages over time



Back

Appendix: Nesting structure



Repair cost governs the comply/non-comply margin; within-nest factors (station availability, outside option) determine the fraud/abandon split

Back

Appendix: Full pollution regression results

	NO (g/kg-fuel)	HC (g/kg-fuel)
Fraud ($\hat{\tau}_i > 0.5$)	2.23	0.45
	(0.03)	(0.01)
Unregistered	0.94	0.42
	(0.03)	(0.02)
Control mean	1.09	0.19
% effect (fraud)	204.6 %	239.2 %
% effect (unreg.)	86.5 %	223.6 %
Observations	1884938	

Preferred specification: make, model-year, capture-date FEs + driving controls (speed, acceleration, VSP). SEs two-way clustered by vehicle and capture date. See paper Table 3 for all specifications.

Back

u